

Rank and nullity

If A is an $m \times n$ matrix then the rank of A is the dimension of the column space of A . The nullity of A is the dimension of the nullspace of A .

Theorem

For an $m \times n$ matrix A , then

$$\underline{\text{rank}(A)} + \underline{\text{nullity}(A)} = n.$$

Moreover, the rank of A is the dimension of the row space of A .

$$\begin{aligned}\text{rank}(A) &= \dim \text{ of the column space of } A \\ &= \dim. \text{ of the row space of } A \\ &= \# \text{ of pivot columns in } \text{rref}(A).\end{aligned}$$

column space of $A \subseteq \mathbb{R}^m$, row space of $A \subseteq \mathbb{R}^n$

nullity of A = dim. of the nullspace of A
i.e. the set of x s.t. $Ax = 0$
 $\subseteq \mathbb{R}^n$

= # of free variables in the solution
of $Ax = 0$.

Ex: If A is 3×4 then T_A is not 1-1.

$$\text{rank}(A) + \text{nullity}(A) = 4$$

But $\text{rank}(A) \leq 3$ because there are only
3 rows so $\text{nullity}(A) \geq 1$ so T_A is not
1-1.

Ex:

$$A = \begin{pmatrix} \downarrow & & \downarrow & \downarrow & & \\ 1 & 3 & 0 & 0 & -1 & 6 \\ 0 & 0 & 1 & 0 & 7 & 0 \\ 0 & 0 & 0 & 1 & 4 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$\uparrow \qquad \qquad \qquad \uparrow \quad \uparrow$

$$\text{rank}(A) = 3$$

$$\underline{\text{nullity}(A) = 3 = 6 - 3}$$

$$\text{Solve } Ax = 0$$

$$x = \begin{pmatrix} -3x_2 + x_5 - 6x_6 \\ x_2 \\ -7x_5 \\ -4x_5 + 3x_6 \\ x_5 \\ x_6 \end{pmatrix} = x_2 \begin{pmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + x_5 \begin{pmatrix} 1 \\ 0 \\ -7 \\ -4 \\ 1 \\ 0 \end{pmatrix} + x_6 \begin{pmatrix} -6 \\ 0 \\ 0 \\ 3 \\ 0 \\ 1 \end{pmatrix}$$

Eigenvalues and eigenvectors

Definition

Suppose that A is an $n \times n$ matrix and $\lambda \in \mathbb{R}$.

- ① If $u \in \mathbb{R}^n$ with $u \neq 0$ and

$$Au = \lambda u$$

then we say that λ is an eigenvalue of A and $\underline{u}$ is an eigenvector for λ .

- ② If λ is an eigenvalue for A then

$$\{u \in \mathbb{R}^n : Au = \lambda u\}$$

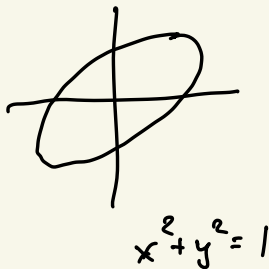
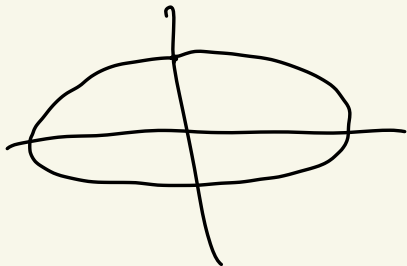
is called the λ -eigenspace for A .

λ -eigenspace is the nullspace of $(A - \lambda I)$.

Ex:

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

$$(x, y)^T A \begin{pmatrix} x \\ y \end{pmatrix} = \underline{3x^2 + 2xy + 3y^2} = 1$$



Ex: $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$, Claim: $\lambda = 2$ and $\underline{4}$
are eigenvalues
for A .

$\lambda = 2$: Solve $Au = 2u$

$$\text{or } \underline{(A - 2I)}u = 0$$

$$(A - 2I)u = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} u = 0$$

$u = \underline{t \begin{pmatrix} 1 \\ -1 \end{pmatrix}}$ for any t . $\Leftarrow$ all
eigenvectors
for $\lambda = 2$
except $t = 0$.

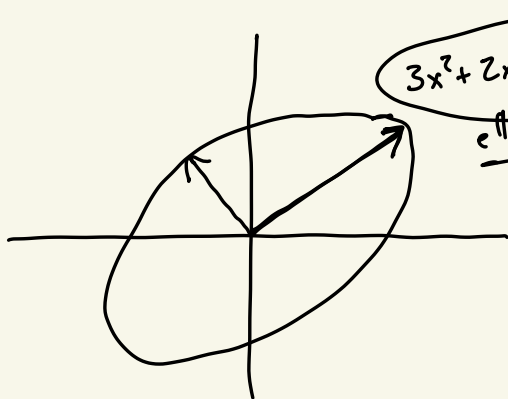
$$\lambda = 4: \quad \text{Solve } Au = 4u$$

$$(A - 4I)u = 0$$

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} u = 0$$

$$u = \underline{t \begin{pmatrix} 1 \\ 1 \end{pmatrix}} \quad \text{for } t \in \mathbb{R}.$$

eigenvectors for $\lambda = 4$.
(except $t = 0$).



$$3x^2 + 2xy + 3y^2 = 1$$

ellipse.

$$\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ and } \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

act as a coordinate system in which this ellipse is in standard form.

Linear independence of eigenvectors

Theorem

Suppose that A is an $n \times n$ matrix and $v_1, \dots, v_k$ are eigenvectors for distinct eigenvalues $\lambda_1, \dots, \lambda_k$. Then $v_1, \dots, v_k$ are linearly independent.

The characteristic equation

Definition

For A , an $n \times n$ matrix, $\det(A - \lambda I_n) = 0$ is called the characteristic equation of A . It is a polynomial of degree n .

- 1 The eigenvalues of A are solutions of the characteristic equation of A .
- 2 The algebraic multiplicity of an eigenvalue λ is the multiplicity of λ as a root of the characteristic equation.
- 3 The geometric multiplicity of λ is the dimension of the λ -eigenspace.
- 4 The geometric multiplicity is always less than or equal to the algebraic multiplicity.

λ is an eigenvalue if $Au = \lambda u$ has a non-trivial solution.
Which means that $(A - \lambda I)u = 0$ has a non-trivial solution.

Ex: $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ and compute $\det(A - \lambda I)$

$$A - \lambda I = \begin{pmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{pmatrix} \quad \text{so}$$

$$\begin{aligned} \det(A - \lambda I) &= (3-\lambda)^2 - 1 \\ &= \lambda^2 - 6\lambda + 8 \end{aligned}$$

$$= \underline{(\lambda - 4)(\lambda - 2)} \leftarrow \begin{array}{l} \text{char.} \\ \text{poly of} \\ A. \end{array}$$

The roots of the char. poly are 2 and 4
which are the eigenvalues.

Ex: $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ char poly is $\det(A - \lambda I)$

$$= \det \left(\begin{pmatrix} 1-\lambda & 1 \\ 0 & 1-\lambda \end{pmatrix} \right)$$
$$= (1-\lambda)^2$$

The only eigenvalue is $\lambda=1$ and it has alg. mult. 2

Check: The 1-eigenspace here is 1-dim.
i.e. has geometric mult 1.

Similarity

Definition

For A and B , two $n \times n$ matrices, we say that A is similar to B if there is an invertible matrix P such that $P^{-1}AP = B$.

Fact: If A and B are similar then they have the same characteristic polynomial.